

GIT-HOL-4

Merge Conflict Resolution

Objective:

- Resolve merge conflicts during branch merging.

Prerequisites:

- Completed HOL 3.

Procedure:

1. Create GitWork branch.
2. Create hello.xml and commit.
3. Switch to main and create hello.xml with different content.
4. Commit the changes.
5. Merge GitWork into main to generate conflict.
6. Resolve the conflict manually.
7. Commit the resolved file.
8. Update .gitignore and delete GitWork.

Commands:

```
git checkout -b GitWork
echo "<message>Hello from GitWork</message>" > hello.xml
git add hello.xml
git commit -m "Added hello.xml in GitWork"
git checkout main
echo "<message>Hello from Main Branch</message>" > hello.xml
git add hello.xml
git commit -m "Added hello.xml in main"
git merge GitWork
git add hello.xml
git commit -m "Resolved merge conflict"
git branch -d GitWork
```

Output:

```
MINGW64/c/Users/tavva/GitDemo
taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (main)
$ git checkout -b Gitwork
Switched to a new branch 'Gitwork'

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (Gitwork)
$ git branch
* Gitwork
  main

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (Gitwork)
$ echo "<message>Hello from Gitwork</message>" > hello.xml

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (Gitwork)
$ cat hello.xml
<message>Hello from Gitwork</message>

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (Gitwork)
$ git add hello.xml
$ git commit -m "Added hello.xml in Gitwork"
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it
[Gitwork 253826e] Added hello.xml in Gitwork
1 file changed, 1 insertion(+)
create mode 100644 hello.xml

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (Gitwork)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (main)
$ echo "<message>Hello from Main Branch</message>" > hello.xml

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (main)
$ cat hello.xml
<message>Hello from Main Branch</message>

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (main)
$ git add hello.xml
$ git commit -m "Added hello.xml in main"
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it
[main fc7aabc] Added hello.xml in main
1 file changed, 1 insertion(+)
create mode 100644 hello.xml

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (main)
$ git log --oneline --graph --decorate --all
* fc7aabc (HEAD -> main) Added hello.xml in main
* 253826e (Gitwork) Added hello.xml in Gitwork
/
* 37b6882 (origin/main) Added branch.txt in GitNewBranch
* 4e82564 Added gitignore file
* f0754c9 Added welcome file

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (main)
```

```
MINGW64/c/Users/tavva/GitDemo
taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (main)
$ git diff main Gitwork
diff --git a/hello.xml b/hello.xml
index 064b23b..4f3217 100644
--- a/hello.xml
+++ b/hello.xml
@@ -1,1 @@
- <message>Hello from Main Branch</message>
+ <message>Hello from Gitwork</message>

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (main)
$ git merge Gitwork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (main[MERGING])
$ notepad hello.xml

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (main[MERGING])
$ git add hello.xml
$ git commit -m "Resolved merge conflict"
[main 07a9978] Resolved merge conflict

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (main)
$ notepad .gitignore

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (main)
$ git add .gitignore
$ git commit -m "Updated .gitignore"
[main 4e4d699] Updated .gitignore
1 file changed, 2 insertions(+), 1 deletion(-)

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (main)
$ git branch
* Gitwork
  main

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (main)
$ git branch -d Gitwork
Deleted branch Gitwork (was 253826e).

taavva@LAPTOP-304UFCD4 MINGW64 ~/GitDemo (main)
$ git log --oneline --graph --decorate
* 4e4d699 (HEAD -> main) Updated .gitignore
* 07a9978 Resolved merge conflict
/
* 253826e Added hello.xml in Gitwork
* fc7aabc Added hello.xml in main
/
* 37b6882 (origin/main) Added branch.txt in GitNewBranch
```

Result:

Merge conflict was resolved successfully and changes were committed.